Hyperbolic Functions: Differentiation

Questions 1–49

Question 1. Show, by writing $\sinh x$, $\cosh x$ and $\tanh x$ in terms of exponentials, that:

a) $\dfrac{d}{dx}(\sinh x) = \cosh x$

b) $\dfrac{d}{dx}(\cosh x) = \sinh x$

c) $\dfrac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$

Prerequisite knowledge for Q2: Hyperbolic Identities

Question 2. It is given that $\lim\limits_{h\to 0} \dfrac{\sinh h}{h} = 1$ and $\lim\limits_{h\to 0} \dfrac{\cosh h - 1}{h} = 0$. Show, by first principles, that:

a) $\dfrac{d}{dx}(\sinh x) = \cosh x$

b) $\dfrac{d}{dx}(\cosh x) = \sinh x$

The following questions use the derivatives of $\sinh kx$, $\cosh kx$ and $\tanh kx$ and require no further Chain, Product or Quotient Rule.

Question 3. Differentiate each of the following:

a) $y = \sinh 2x$
b) $y = \cosh 3x$
c) $y = \tanh 6x$
d) $y = \sinh \frac{1}{4}x$
e) $y = \cosh \frac{x}{3}$
f) $y = \tanh\left(-\frac{x}{5}\right)$

Question 4. Differentiate each of the following:

a) $y = 3\sinh 4x - 6\tanh 3x$
b) $y = 5\cosh \frac{x}{10} + \frac{1}{2}\tanh \frac{x}{4}$

Question 5. Find the exact gradient of the curve $y = 3\tanh\frac{1}{2}x$ at $\left(\ln 16, \frac{45}{17}\right)$.

Question 6. Find the equation of the tangent to the curve $y = 3\sinh x$ when $x = \ln 6$, writing your answer in the form $ax + by = c$ where a, b and c are constants.

Question 7. Find the equation of the tangent to the curve $y = 5\sinh x - \cosh x$ when $x = \ln 2$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

Question 8. Find the equation of the tangent to the curve $y = 3\cosh 2x - 4\sinh 3x$ when $x = \ln \frac{1}{3}$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

Question 9. Find the equation of the tangent to the curve $y = \tanh \frac{x}{2}$ when $x = \ln 4$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

Question 10. Find the equation of the normal to the curve $y = 8 + 6\tanh \frac{1}{3}x - 4\tanh \frac{2}{3}x$ when $x = 0$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

Question 11. The line l_1 is the normal to the curve $y = 12\sinh\frac{x}{3}$ when $x = \ln 8$, and the line l_2 is the normal to the curve $y = 6\cosh\frac{x}{4}$ when $x = \ln 16$. Show that the lines l_1 and l_2 intersect when $x = a\ln 2 + b$, where a and b are rational numbers to be found.

Question 12. Show that the curve with equation $y = \sinh x + \tanh x$ is always increasing.

Question 13. It is given that the curve with equation $y = \sinh x + 2\cosh x$ has a single stationary point. Find the exact coordinates of the stationary point. **Prerequisite knowledge for Q14: Hyperbolic Double-Angle Formulae**

Question 14. Show that the curve with equation $y = 9\cosh x + 3\sinh 2x$ has no point where the gradient is 1.

Question 15. It is given that $f(x) = p\sinh x - q\cosh x$ where p and q are positive constants. Show that the curve $y = f(x)$ has no stationary points when $p \geq q$.

Question 16. It is given that the curve with equation $y = \sinh x - 8\tanh x$ has two stationary points. Find the exact coordinates of both stationary points.

Question 17. Show that the curve $y = 5\cosh x + \sinh x + x$ has a single stationary point.

Question 18. Determine whether the curve with equation $y = \cosh 3x - 2\sinh x$ is convex (concave upwards) or concave (concave downwards) when $x = \ln 2$.

Question 19. Show that the curve $y = 4\sinh 2x - 3\cosh 2x$ has a single point of inflection and find its exact coordinates. **The following questions use the derivatives of** $\sinh kx$**,** $\cosh kx$ **and** $\tanh kx$ **and use any differentiation knowledge from A-Level Maths.**

Question 20. Differentiate each of the following:

a) $y = \sinh(\sqrt{x})$

b) $y = \cosh\left(\frac{1}{\sqrt{x}}\right)$

c) $y = \tanh\left(\frac{x+1}{2\sqrt{x}}\right)$

Question 21. Differentiate each of the following:

a) $y = \sinh^2 x$
b) $y = \cosh^2 x$
c) $y = \sinh^3 x$
d) $y = \cosh^4 x$
e) $y = \tanh^2 x$
f) $y = \tanh^5 x$

Question 22. Show that the curve $y = 3\cosh^2 x - \sinh x$ has a single stationary point.

Question 23. It is given that $f(x) = \cosh^2 2x - p\cosh 2x$, where p is a constant. Find the minimum value of p such that $y = f(x)$ has more than one stationary point.

Question 24. Differentiate each of the following:

a) $y = \sinh(\cosh x)$

b) $y = \cosh(\sinh x)$

c) $y = \tanh(\tanh x)$

Question 25. Show that:

a) $\dfrac{d}{dx}(\ln(\sinh x)) = \coth x$

b) $\dfrac{d}{dx}(\ln(\cosh x)) = \tanh x$

Prerequisite knowledge for Q26: Hyperbolic Double-Angle Formulae

Question 26. Show that $\dfrac{d}{dx}\left(\dfrac{1}{2}\ln(\tanh x)\right) = \operatorname{cosech} 2x$.

Question 27. Differentiate each of the following:

a) $y = \sinh x \cosh x$

b) $y = \sinh 3x \cosh 4x$

c) $y = x \sinh x$

d) $y = x^2 \cosh x$

e) $y = (x+1) \sinh 2x$

f) $y = (1 - x^2) \cosh \frac{x}{2}$

Question 28. Find the exact gradient of the curve $y = \sinh 2x \tanh 3x$ at $\left(\ln 2, \frac{189}{104}\right)$.

Question 29. Find the equation of the tangent to the curve $y = (x^2 + x - 6)\cosh x$ when $x = 0$, writing your answer in the form $y = ax + b$ where a and b are integers.

Question 30. Find the equation of the normal to the curve $y = \sinh(12x - \ln 2)\cosh(3x - \ln 3)$ when $x = 0$, writing your answer in the form $ax + by = c$ where a, b and c are integers.

Question 31. Show that the curve with equation $y = \sinh x \cosh x$ is always increasing.

Question 32. It is given that the curve with equation $y = (\cosh x - 3)\cosh x$ has three stationary points. Find the exact coordinates of all three stationary points.

Question 33. Determine whether the curve with equation $y = \sinh\frac{x}{2}\left(2 - \sinh\frac{x}{3}\right)$ is convex (concave upwards) or concave (concave downwards) when $x = \ln 64$.

Question 34. Show that the curve $y = (\sinh x - 6)\sinh x$ has two points of inflection and find their exact coordinates.

Question 35. By writing $\tanh x$ as $\dfrac{\sinh x}{\cosh x}$, use the quotient rule to show that $\dfrac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$.

Question 36. Differentiate each of the following:

a) $y = \dfrac{\sinh x}{\cosh 2x}$

b) $y = \dfrac{\cosh x}{\sinh 2x}$

c) $y = \dfrac{x}{\sinh x}$

d) $y = \dfrac{\cosh x}{x}$

e) $y = \dfrac{x+1}{\sinh 2x}$

f) $y = \dfrac{\cosh 3x}{x^2+1}$

Question 37. Find the exact gradient of the curve $y = \dfrac{\sinh 3x}{\sinh x}$ at $\left(\ln 2, \frac{21}{4}\right)$.

Question 38. Find the equation of the tangent to the curve $y = \dfrac{x + \cosh 3x}{1 - \sinh 2x}$ when $x = 0$, writing your answer in the form $y = ax + b$ where a and b are integers.

Question 39. Find the equation of the normal to the curve $y = \dfrac{\tanh 3x + 6}{\cosh 2x + 3}$ when $x = 0$, writing your answer in the form $ax + by = c$ where a, b and c are integers.

Question 40. It is given that $f(x) = \dfrac{x+1}{\sinh x}$.

a) Show that the x-coordinates of the stationary points of the curve $y = f(x)$ satisfy $\tanh x = x + 1$.

b) By sketching $y = \tanh x$ and $y = x + 1$ on the same axes, determine the number of stationary points the curve $y = f(x)$ has.

Question 41. It is given that the curve with equation $y = \dfrac{\sinh x + 1}{\tanh x}$ has a single stationary point. Find the exact coordinates of the stationary point.

Question 42. It is given that the curve with equation $y = \dfrac{\cosh x}{4 - \sinh x}$ has a gradient of $\frac{1}{16}$ at two points. One of these points is $x = 0$. Find the exact x-coordinate of the other point.

Question 43. Show that the curve $y = \dfrac{4}{\cosh x}$ has two points of inflection and find their exact coordinates.
The following questions use the derivatives of $\operatorname{sech} x$, $\operatorname{cosech} x$ **and** $\coth x$ **and use any differentiation knowledge from A-Level Maths.**

Question 44. By writing $\coth x$ as $\dfrac{\cosh x}{\sinh x}$, use the quotient rule to show that $\dfrac{d}{dx}(\coth x) = -\operatorname{cosech}^2 x$.

Question 45. By writing $\operatorname{cosech} x$ as $\dfrac{1}{\sinh x}$, use the quotient rule to show that

$$\frac{d}{dx}(\operatorname{cosech} x) = -\coth x \operatorname{cosech} x.$$

Question 46. By writing $\operatorname{sech} x$ as $\dfrac{1}{\cosh x}$, use the quotient rule to show that $\dfrac{d}{dx}(\operatorname{sech} x) = -\tanh x \operatorname{sech} x$.

Question 47. Show that the curve with equation $y = 2x + \operatorname{sech} x$ has two points of inflection. **Prerequisite knowledge for Q48: Hyperbolic Identities**

Question 48. Show that $\dfrac{d}{dx}\left(\ln \tanh \dfrac{x}{2}\right) = \operatorname{cosech} x$. **Prerequisite knowledge for Q49: Differentiation of Inverse Trigonometric Functions**

Question 49. Given that $y = \arctan(4e^{5x})$, show that $\dfrac{dy}{dx} = \dfrac{20}{p\sinh 5x + q\cosh 5x}$ where p and q are integers to be found.